

CS449-CS549 Midterm #2

I. QUESTION (8 POINTS)

- Take the derivative of the sigmoid function

$$F(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

and prove that $F'(x) = F(x)(1 - F(x))$

- Take the derivative of the hyperbolic tangent function

$$G(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (2)$$

and prove that $G'(x) = 1 - G(x)^2$

II. QUESTION (12 POINTS)

Given the following documents:

Doc1: ali topu at

Doc2: ali ayşeye çiçek al

Doc3: ali topu sev

Doc4: ayşe topu at

Doc5: ali ayşeye topu at

- Calculate the term-document matrix.
- Calculate the term-term matrix for a word window of ± 1 .
- Using the term-term matrix in b., calculate the cosine similarity between words ali and ayşe.

III. QUESTION (12 POINTS)

Derive the update equation for the logistic regression for the cross-entropy loss function, where the sigmoid is changed with the hyperbolic tangent function

$$G(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3)$$

Remember the cross-entropy loss for sigmoid function is defined as:

$$L(w, b) = -[y \log \sigma(wx+b) + (1-y) \log(1-\sigma(wx+b))] \quad (4)$$

IV. QUESTION (8 POINTS)

Given the following contingency table, calculate the precision, recall, F-measure and accuracy scores.

	Gold positive	Gold negative
System positive	70	30
System negative	40	60

V. QUESTION (10 POINTS)

Given the following grammar L_1 below:

S	$\rightarrow$	$NP VP$
NP	$\rightarrow$	$Pronoun$
NP	$\rightarrow$	$ProperNoun$
NP	$\rightarrow$	$Det Nominal$
$Nominal$	$\rightarrow$	$Nominal Noun$
$Nominal$	$\rightarrow$	$Noun$
VP	$\rightarrow$	$Verb$
VP	$\rightarrow$	$Verb NP$
VP	$\rightarrow$	$Verb NP PP$
VP	$\rightarrow$	$Verb PP$
VP	$\rightarrow$	$VP PP$
PP	$\rightarrow$	$Prep NP$
$Noun$	$\rightarrow$	$flights, breeze, trip, morning$
$Verb$	$\rightarrow$	$is, prefer, like, need, want, fly$
$Pronoun$	$\rightarrow$	me, I, you, it
$ProperNoun$	$\rightarrow$	$Alaska, Baltimore, Chicago, United$
Det	$\rightarrow$	$the, a, an, this, these, that$
$Prep$	$\rightarrow$	$from, to, on, near$

- Construct the parse tree for "I like the morning trip"
- Extract grammar rules and lexical entries for the parse tree in a.
- Construct an ambiguous sentence for the grammar L_1 .

VI. QUESTION (9 POINTS)

Given the following sentences, annotate the corresponding universal dependency relations on them.

Sentence	Relation
United canceled the flight	NSUBJ
United diverted the flight to Miami	DOBJ
We booked her the flight to Miami	IOBJ
We took the morning flight	NMOD
Book the cheapest flight	AMOD
JetBlue canceled 1000 flights	NUMMOD
The flight was canceled	DET
Book the flight through Houston	CASE
We flew Denver and Houston	CC, CONJ

VII. QUESTION (11 POINTS)

Given the following grammar L_2 in CNF form:

- $$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Noun Noun \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flight, trip, super, morning \\
 Verb &\rightarrow is \\
 Det &\rightarrow the, this \\
 Prep &\rightarrow on
 \end{aligned}$$

Parse the sentence "The morning flight is on this super trip" using CKY parsing. Show your work.

VIII. QUESTION (10 POINTS)

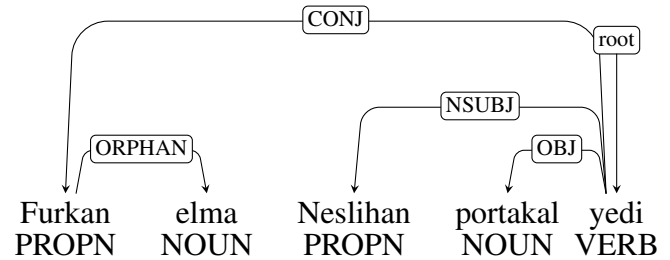
Given the following grammar L_1 below:

- $$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Pronoun \\
 NP &\rightarrow ProperNoun \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Nominal Noun \\
 Nominal &\rightarrow Noun \\
 VP &\rightarrow Verb \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb NP PP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flights, breeze, trip, morning \\
 Verb &\rightarrow is, prefer, like, need, want, fly
 \end{aligned}$$

- $$\begin{aligned}
 Pronoun &\rightarrow me, I, you, it \\
 ProperNoun &\rightarrow Alaska, Baltimore, Chicago, United \\
 Det &\rightarrow the, a, an, this, these, that \\
 Prep &\rightarrow from, to, on, near
 \end{aligned}$$

Convert the grammar into CNF form.

IX. QUESTION (10 POINTS)



Construct the trace of the given transition-based parse above. Show stack, word list, action and relation added at each iteration. Assume that arc-standard parsing has been used.

X. QUESTION (10 POINTS)

Draw a single input, single hidden layer (with a single hidden node) neural network (multilayer perceptron) for two-class classification problem. Derive the update equation for

- the weight between the input unit and hidden layer
- the weight between the hidden unit and the output.

Do not forget the bias unit.